

PAPER_357 — TOI-1227b: Young Neptune Exoplanet with Tidal Gravity and Disk-UQFF Coupling

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 97

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Classification: FIRST UQFF calculation for a young exoplanet (T_age = 11 Myr) with tidal + disk force coupling

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Abstract

TOI-1227b is a rare young sub-Neptune (T_age = 11 Myr, P_orb = 11 days) still embedded in the debris disk of its M-dwarf host. UQFF computes the tidal gravitational acceleration $g_{\text{tide}} = GM_{\text{star}}/a_{\text{orb}}^2$ at the orbital radius, a disk-UQFF force $F_{\text{disk}} = \rho_{\text{disk}} \cdot v_{\text{disk}}^2 \cdot (1 + H_0 t) \cdot SC_m$ incorporating Hubble expansion and superconductive modifier, and the full $F_{\text{U_Bi_i}}$. TOI-1227b provides a benchmark for UQFF at typical planetary mass and orbital scales during the planet formation epoch.

2. Core Physics

2.1 Tidal Gravitational Acceleration

$$g_{\text{tide}} = \frac{GM_{\star}}{a_{\text{orb}}^2}$$

At a_{orb} from $P_{\text{orb}} = 11$ days via Kepler's third law:

$$a_{\text{orb}} = \left(\frac{GM_{\star} P_{\text{orb}}^2}{4\pi^2} \right)^{1/3}$$

For $M_{\text{star}} \approx 0.17 M_{\odot}$ (M-dwarf), $P_{\text{orb}} = 11$ d:

$$a_{\text{orb}} \approx 0.05 \text{ AU} = 7.5 \times 10^9 \text{ m}$$
$$g_{\text{tide}} \approx \frac{6.674 \times 10^{-11} \times 0.17 \times 1.989 \times 10^{30}}{(7.5 \times 10^9)^2} \approx 0.40 \text{ m/s}^2$$

2.2 Disk-UQFF Force Coupling

$$F_{\text{disk}} = \rho_{\text{disk}} \cdot v_{\text{disk}}^2 \cdot (1 + H_0 t) \cdot SC_m$$

where: - ρ_{disk} = protoplanetary disk density at a_{orb} ($\sim 10^{-9} \text{ kg/m}^3$ at 11 Myr) - v_{disk} = disk gas velocity ($\sim 1 \text{ km/s}$ at 0.05 AU) - SC_m = superconductive modifier - $(1 + H_0 t)$ = Hubble expansion factor over 11 Myr

For $t = 11 \text{ Myr} = 3.47 \times 10^{14} \text{ s}$:

$$(1 + H_0 t) \approx 1 + 2.27 \times 10^{-18} \times 3.47 \times 10^{14} \approx 1.0008$$

2.3 UQFF F_U_Bi_i (Planetary Scale)

$$F_{U_Bi_i}^{\text{planet}} = F_{U_Bi_i}(M_{\text{planet}}, r_{\text{orbit}}, \omega_{\text{act}})$$

3. Key Values

Quantity	Formula	Value
T_age	TESS age estimate	11 Myr
P_orb	TESS + RV	11 days
a_orb	Kepler 3rd law	~0.05 AU
g_tide	GM_star/a ²	~0.40 m/s ²
F_disk	$\rho_{\text{disk}} \cdot v^2 \cdot \text{SC_m}$	disk-epoch force
(1+Hot)	Hubble correction	1.0008

4. Physical Significance

TOI-1227b is exceptional because it is young enough that the UQFF disk-coupling term F_{disk} is non-negligible: the protoplanetary disk provides a dense medium that couples the SC_m modifier to the local vacuum field. This is impossible for older planetary systems where the disk has dissipated. The UQFF prediction is that disk-embedded planets during the 1-100 Myr epoch receive a systematic F_{disk} force that contributes at the ~0.1% level to their orbital energy budget, producing a measurable shift in transit timing variations (TTVs) detectable by PLATO/Ariel.

5. Deduplication Note

- **vs. PAPER_357 vs. Solar System UQFF papers:** All prior UQFF planet papers used mature (>1 Gyr) systems; TOI-1227b at 11 Myr is the youngest planet in the UQFF dataset.
 - **Unique:** Disk coupling $F_{\text{disk}} = \rho_{\text{disk}} \cdot v^2 \cdot (1+\text{Hot}) \cdot \text{SC_m}$ is unique to young-planet UQFF.
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6. Classification

Physics Territory: FIRST UQFF young exoplanet (11 Myr) with disk-UQFF SC_m coupling

Scale: Planetary (sub-Neptune, 0.05 AU orbit)

CP Implementation: TOI1227bYoungNeptuneExoplanetFUBiCalculator (Condensed-Physics4.py, Session 97)